

CptS 355 Study Group — Week 8

Scribe: Claire **Attended:** Jason, Sarah, Arvind, Claire (all four)

Met Wednesday at 7:00pm on the ground floor of SEH, for about 60 minutes.

- Went back over the lambda calculus part of Tuesday's lecture.
- How we've understood free variables: free in a body if no enclosing lambda binds it.
- Marked every variable free or bound in four small expressions. Sarah caught one.
- A name can be free in the inner lambda and bound in the outer one.
- A closure is the lambda plus the bindings for its free variables.
- Returned a lambda out of a Scheme function and called it after the outer call. Tracked what happened to the free variable and checked it against beta reduction by hand.
- We compared our solutions on the counting-terms exercise and agreed Jason had an elegant way of breaking the problem into components that kept things simple.
- Wrote down Church 0, 1, and 2, and what the two arguments are for.
- Could not get Church addition right. Ours gave the wrong numeral. Arvind is redoing it before next week.

The example we spent the most time on:

```
(define (make-adder n) (lambda (x) (+ x n)))  
(define add5 (make-adder 5))    ; n is free in the inner lambda
```

By the time `add5` runs, `n` is out of scope, so the closure is carrying it.